Shivam Upadhyay

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Degree	University/Institute	Year	CGPA/Marks(%)
M.Tech (MDI)	IIT Hyderabad	2025	9.37
B.Tech (Biotechnology)	Bundelkhand University, Jhansi, UP	2023	8.15
XII (CBSE)	S.M.P.K.D Shikshan Sansthan, Firozabad, UP	2017	82%
X (CBSE)	S.M.P.K.D Shikshan Sansthan, Firozabad, UP	2015	9.80

SKILLS

Programming Languages: Python, SQL
DevOps: Google Cloud Platform
Others: Git, Pandas, Docker, Numpy, Matplotlib, Raspberry Pi, Arduino
Circuit Design and Analysis, Prototyping and Fabrication
Biotech and Laboratory skills

EXPERIENCE

Research Fellow – PLRI Lab, TU Braunschweig, Germany | Link *Sep'24 - Mar'25*

- Developed an ISAN-based data exchange framework for emergency medical systems integrating IoT and hospital ICT systems.
- Simulated real-time ECG and PPG signals at 500 Hz using Python on Raspberry Pi and transmitted to the ISAN server.
- Designed a communication protocol between Curing System (CS) and Responding System (RS), measuring system latency.

Intern at QC and QA, Venus Remedies Limited, Haryana *Jun'22-Jul'22*

- Gained hands-on experience in pharmaceutical quality control and assurance processes, including product testing and validation.
- Assisted in ensuring regulatory compliance, documentation accuracy, and audit readiness.
- Collaborated with cross-functional teams to support process optimization and continuous improvement in production quality.

PROJECTS

(Build Funded) Design and Validation of Multimodal Flexible Wearable Band for Real-Time Vital Sign Monitoring in Ambulance *Mar'24- May'25 Dr. Nagarajan Ganapathy*

- Designed and developed a flexible wearable band to monitor ECG, PPG, temperature, and GSR in real-time during ambulance transport.
- Integrated biomedical sensors and wireless modules for continuous data acquisition and transmission to emergency response systems.
- Enhanced pre-hospital care by enabling real-time vitals monitoring, supporting faster clinical decision-making and improved patient outcomes.

Production of Bacterial Biomass as Bio-Fertilizer to Increase Soil Fertility *Feb'22 - May'23 Dr. Brijendra Shukla*

- Cultured *Rhizobium leguminosarum* to produce bio-fertilizer and evaluated its impact on plant growth across different soil types.
- Conducted field trials to assess effectiveness in enhancing soil fertility and promoting sustainable agriculture.
- Analyzed sustainability, scalability, and future potential of microbial-based bio-fertilizer solutions.

ACHIEVEMENTS

- Recipient of **DAAD-KOSPIE** Scholarship for academic and research exchange at Technische Universität Braunschweig, Germany.
- LIC HFL Scholastic Achievement Award Recognized for outstanding academic excellence.
- Competitive Exam Achievements Qualified **GATE** (BT & XL) in 2022 & 2023, **GAT-B** (2023), and **BITP** (2023).

CERTIFICATIONS

- Google Cloud for Beginners. Start Cloud Computing with GCP *July'25*
- 1st Year Academic Excellence Award in MTech Program IIT Hyderabad *Apr' 25*
- Data Analytics | **GFG** *May'24*

PATENTS & PUBLICATIONS

- Transmission of Vital Data in Emergencies using the International Standard Accident Number | **EFMI STC**
- Assessment of Single-Arm, Single-Lead ECG using 3D-Printed Dry Electrodes in Dynamic Conditions | **EMBC**
- **Patents**: System and Method for Real-Time ECG Monitoring Using a Single-Arm, Single-Lead Dry Electrode Configuration for Emergency and Ambulatory Care